

ENGINEERING MECHANICS

I Year II Semester: MECHANICAL ENGINEERING & AERONAUTICAL ENGINEERING								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME05	ESC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OUTCOMES:								
At the end of the course the student should be able to:								
1. Evaluate different system of forces acting on rigid body under static equilibrium.								
2. Analyse the frictional forces existing at different contact surfaces of the rigid body under static equilibrium.								
3. Determine the centre of gravity and moment of inertia of simple and composite bodies.								
4. Analyse linear and rotary motion of particle or rigid bodies using Kinematic & Kinetic Equations.								
5. Evaluate solutions for mechanics problems by using virtual work method and also able to distinguish basics of mechanical vibrations.								
UNIT-I	INTRODUCTION TO ENGINEERING MECHANICS & EQUILIBRIUM OF PLANAR SYSTEM OF FORCES							
Introduction to engineering mechanics: Basic concepts, Classification of a force system, Parallelogram law, Resultant of coplanar concurrent force system, Moment of a force, its applications, Varignon's theorem, Resultant of coplanar non-concurrent force systems, Couples.								
Equilibrium of Planar System of forces: Equilibrium, Conditions of equilibrium, free body diagrams, Lami's theorem.								
UNIT-II	FRICTION							
Friction: Introduction, types of friction, laws of friction, coefficient of friction, angle of friction, angle of repose, analysis of bodies on rough horizontal, inclined planes, wedge friction, ladder friction and screw friction								
UNIT-III	CENTROID, CENTRE OF GRAVITY & MOMENT OF INERTIA							
Centroid & Centre of gravity: Introduction, centroids & centre of gravity of simple figures (from first principles), centroid of composite sections, centre of gravity of composite solids, theorems of pappus & Guldinus.								
Moment of Inertia: Area moment of inertia of plane sections (from first principles), Transfer theorems, Moment of inertia of standard sections and composite sections. Mass moment of inertia of rectangular plate, circular plate. cone, cylinder and sphere.								
UNIT-IV	KINEMATICS & KINETICS							
Kinematics of a particle: rectilinear motion, curvilinear motion, Rotary motion.								
Kinetics of rigid bodies: Analysis of connecting bodies, kinetics of rigid body rotation, D' Alembert's principle, work-energy theorem, impulse-momentum principle, their applications								

UNIT-V	VIRTUAL WORK & MECHANICAL VIBRATIONS
Virtual work: Concept of virtual work, Principle of virtual work, its applications.	
Mechanical Vibrations: Basic terminology, free and forced vibrations, resonance, simple harmonic motion, simple, compound and torsional pendulums.	
Text Books:	
1. Reddy Vijay Kumar and K. Suresh Kumar, Singer's Engineering Mechanics 2. Engineering Mechanics/Timoshenko and D.H. Young, Mc Graw Hill Book Company 3. Bansal R.K, A Text book of Engineering Mechanics, Laxmi Publications	
Reference Books:	
1. Engineering Mechanics/Irving Shames/Prentice Hall 2. Engineering Mechanics/N.H Dubey/Tata Mc Graw Hill 3. A Text book of Engineering Mechanics/ R S Khurmi / S Chand	